

Memory-First Web Agent

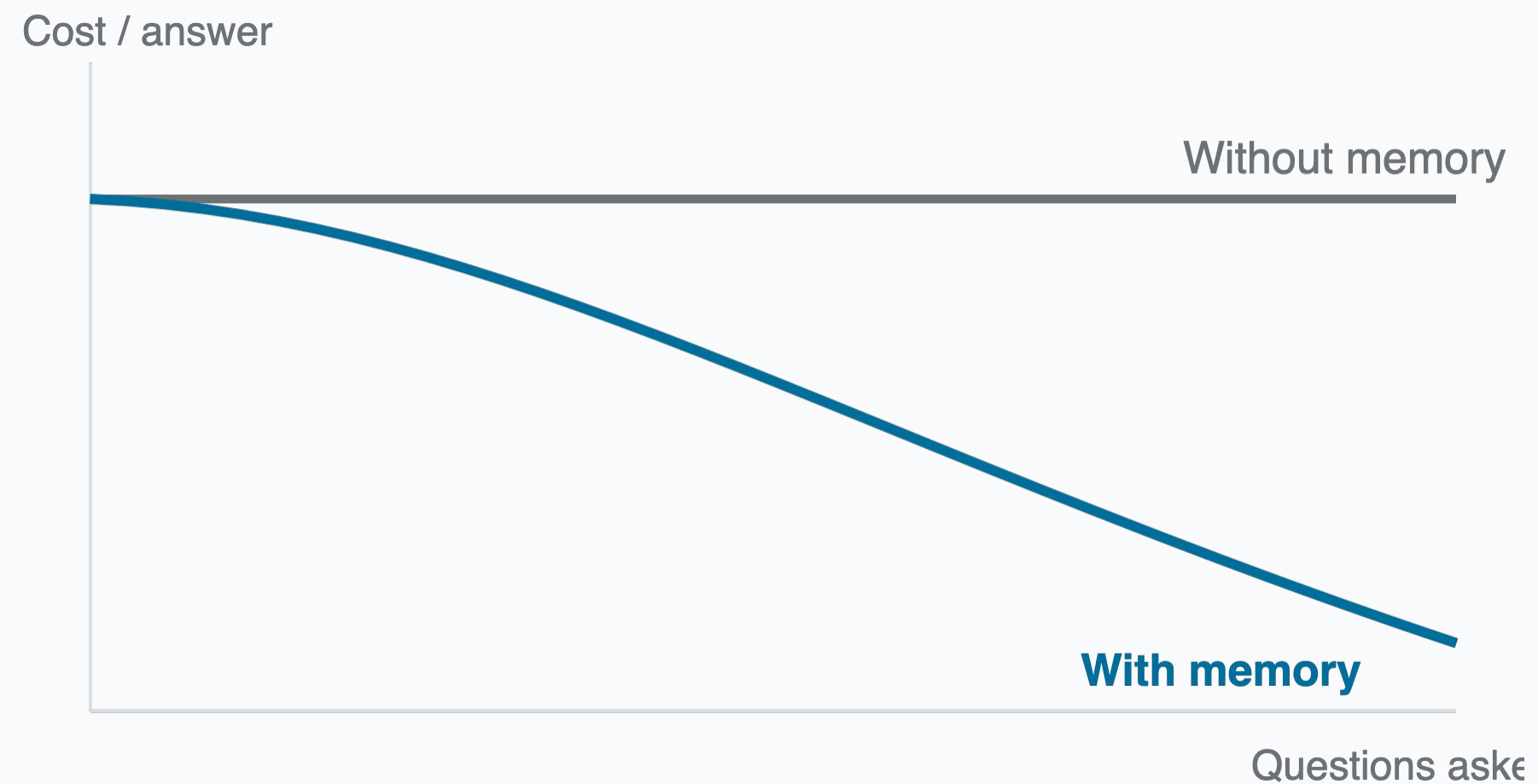
Grounded answers with compounding economics

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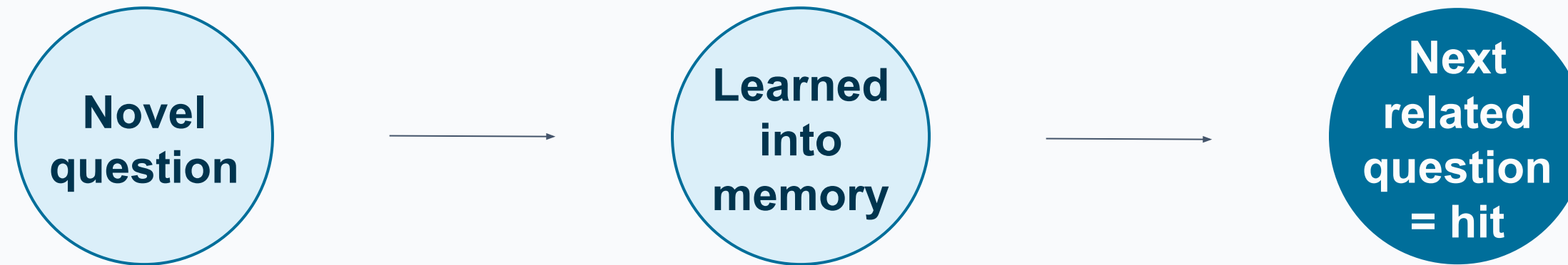
Solution Architect

Every question re-pays the same search and token costs

A plain web-search agent has no memory. Every question — even one asked yesterday — pays full search-API and token cost again. Cost scales linearly with adoption, with no learning effect.



Memory first, web on a miss, learn what you find



Every answer it learns makes the next one cheaper and faster.

Repeat answers cost 100× less — and adoption makes it cheaper

~100×

cheaper for a repeat answer
served from the cache

−47%

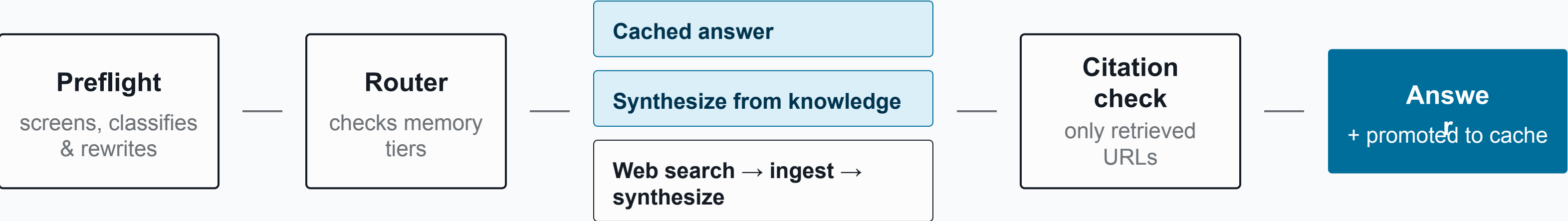
total cost at a 50% memory hit
rate

1–3s vs **8–15s**

memory hit vs. a novel question
answered live

The same hit rate that cuts cost also raises the sustainable throughput ceiling against model quota — one lever, two payoffs.

One turn, start to finish



Two memory tiers, gated for how they actually match

Answer Cache

question ↔ question similarity

gate 0.85

Serves a near-identical repeat instantly.

Knowledge Base

question ↔ chunk similarity

gate 0.70

Grounds a novel-but-related question from what's already known.

Static

Slow-changing

Volatil

e

Volatile queries always bypass memory — fresh by design, not TTL guesswork.

Retrieved content is treated as untrusted until it's verified

In a memory-first agent, retrieved web content becomes persistent state. Memory is treated as a trust boundary.

- 1 Input pattern screen + classifier catch injection at the door
- 2 Ingest screening quarantines flagged chunks — never rewrites them
- 3 Prompt spotlighting frames retrieved content as untrusted data
- 4 Citation-subset validation enforced in code, not by prompt
- 5 Least privilege: no side-effecting tools, so a successful injection can at most produce a wrong answer

When search fails, the agent says **so**

Labeled degraded answer

Served from borderline
memory, clearly flagged as
such

Honest refusal

No confident guess when there
is nothing grounded to answer
from

Zero silent-wrong-answer paths.

Forgetting a source is one command, not a reindex

```
agent memory forget --url
```



Cascades by provenance through both memory tiers

PII gate keeps personal questions out of shared memory — fails closed

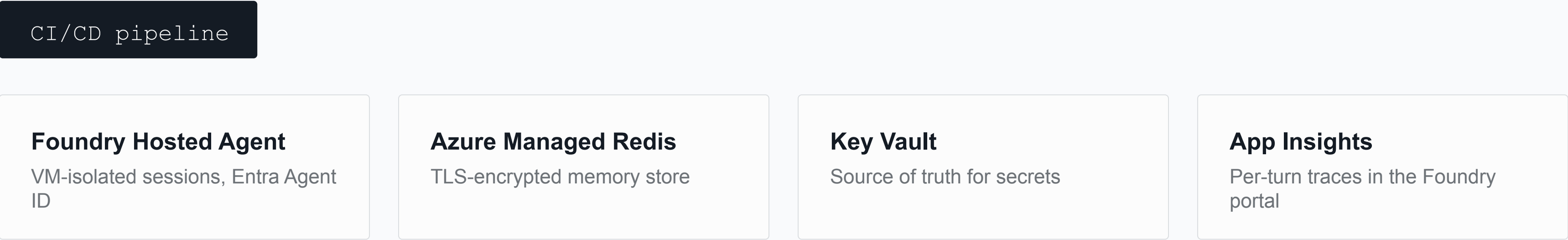
No user accounts; sessions expire after 1 hour

EU data residency (Sweden Central)

Aligned to ISO 27001 Annex A

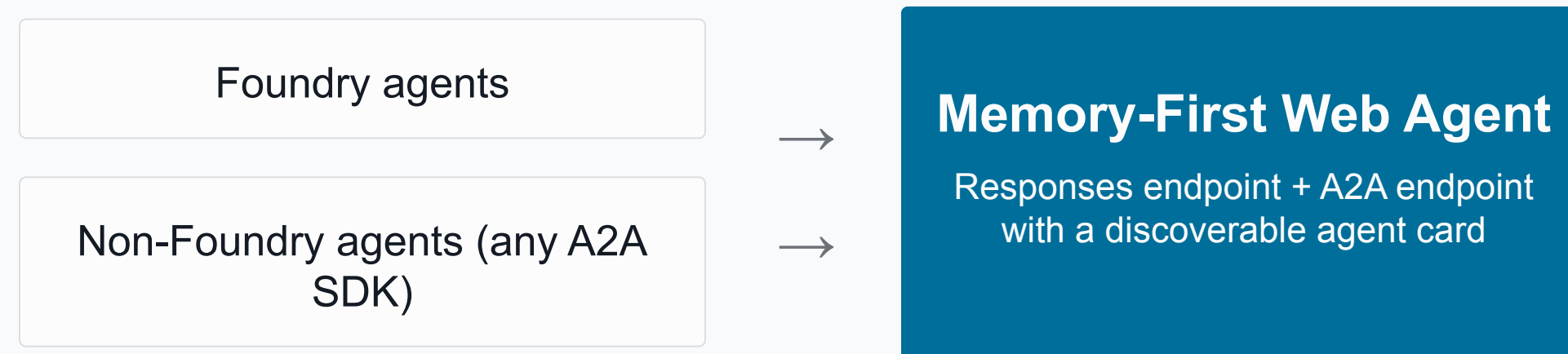
Memory is never the system of record — total loss only degrades the agent to a plain web agent that re-warms through use

Deployed to Azure through CI/CD



OIDC-federated pipeline — no long-lived cloud secrets, provisioned via Bicep IaC.

One agent, two doors into your estate



It deliberately never delegates outward.

Verified in tests, not just claimed

97 automated tests + a deterministic eval suite

14 golden routing cases run against real vector-similarity math

Red-team eval: a poisoned page's injected section is quarantined and never surfaces later

Groundedness measured 4.12 / 5, LLM-judged

Memory hit rate	Cost / 1K turns
0%	\$10.00
30%	−28%
50%	−47%
70%	−66%

One KPI to run this by: memory hit rate.

Roadmap and a 30-day pilot

Near-term

Shave the preflight latency floor on cache hits

Near-term

Calibrate similarity thresholds from live telemetry

Near-term

Optional stronger synthesis model, gated by evals

PROPOSED PILOT

Deploy into your Foundry tenant and measure memory hit rate on real traffic for 30 days.